Rapport de Projet

Oracle Panorama Image Stitching

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Overview

This project builds a fully classical image stitching pipeline combining overlapping photographs into seamless panoramas using SIFT, RANSAC homography estimation, projective warping, and multi-scale blending — no learning-based component is used. Three real datasets were tested: a 2-image coastal sunset (`data3`), a 3-image elevated panorama (`data1`), and a 3-image Lac d'Annecy lakeside scene (`data2`).

Pipeline

Preprocessing

Images are resized to a 1 500 px maximum dimension. CLAHE is applied in the L channel of LAB colour space to locally boost contrast in flat regions (sky, water), significantly increasing keypoint yield in otherwise featureless zones.

Keypoint Detection — SIFT

SIFT builds a Difference-of-Gaussians scale-space pyramid and detects extrema stable across scale and rotation. Each keypoint is described by a **128-d** L2-normalised gradient histogram vector over a 16×16 neighbourhood. We use 5 000–8 000 keypoints with a lowered contrast threshold (0.01–0.02) to handle low-texture regions.

Feature Matching — FLANN + Lowe Ratio Test

FLANN with randomised kd-trees retrieves the 2 nearest neighbours per descriptor. Lowe's ratio test accepts a match only if $d_1 < 0.75\, d_2$, removing ambiguous correspondences before homography estimation.

Homography Estimation — RANSAC

A 3×3 projective homography $\mathbf{H}$ (8 DoF) is estimated via DLT. RANSAC samples 4 correspondences per iteration (10 000 iterations, $\varepsilon = 5$ px inlier threshold). Pairwise transforms are chained to a global canvas frame:

$$H_i^{\text{canvas}} = H_{i-1}^{\text{canvas}} \cdot H_{i \to i-1}$$

Warping & Blending

Each image is warped via `cv2.warpPerspective` onto a shared canvas sized by bounding all projected corners. Two blending modes were compared:

- **Alpha blending:** pixel weights from distance transform to image boundary. Fast, but ghosting artefacts appear near seams.
- **Multi-band (5-level Laplacian pyramid):** blends each frequency band with an appropriately smoothed mask, eliminating ghosting and gracefully handling exposure differences.

Results

Pair	Matches	Inliers	Inlier %	Reproj. err.
data3: 6625 ↔ 6626	505	480	95.0%	0.15 px
data1: 6622 ↔ 6623	281	232	82.6%	1.86 px
data1: 6623 ↔ 6624	226	201	88.9%	1.60 px
data2: 6628 ↔ 6629	353	266	75.4%	1.08 px
data2: 6629 ↔ 6630	1 511	1 467	97.1%	0.89 px

data3 achieves the lowest error (0.15 px) due to clean overlap and simple geometry. **data1** shows higher error (≈1.7 px) from slight perspective drift between hand-held portrait shots.

data2 was the hardest — sparse keypoints on the flat lake required 8 000 features and threshold 0.01; the textured dock area in pair 6629–6630 compensated with 1 467 inliers. Multi-band blending eliminated visible ghosting present in all alpha-blended outputs.

The three panorama outputs are shown below. All images were blended using the multi-band Laplacian pyramid strategy.

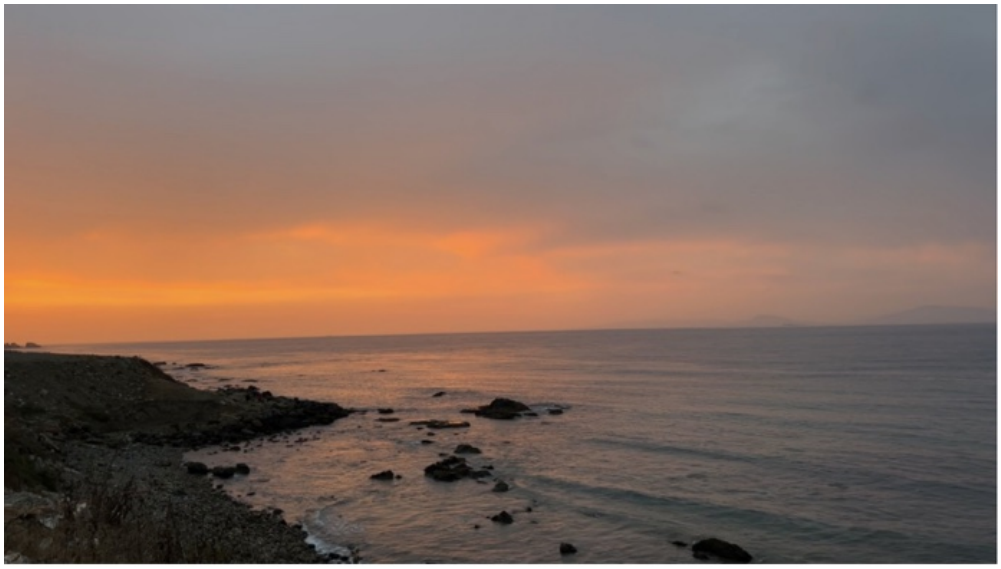
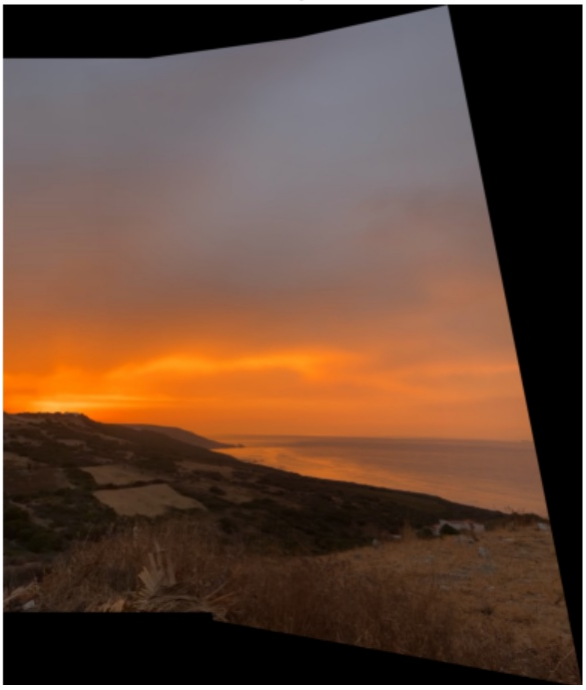


Figure 1: **data3** — Coastal Sunset panorama (2 images, canvas $2\,324 \times 1\,324$ px). Clean alignment with 95% inlier rate and sub-pixel reprojection error.

(a) **data2** — Lac d'Annecy (3 images). Hardest scene due to flat lake surface; CLAHE and higher feature count were essential.



(b) **data1** — Elevated Coastal Panorama (3 images). Portrait-orientation shots introduce slight perspective warp visible at canvas edges.

Figure 2: Multi-band blended panorama outputs for data2 and data1.

Challenges & Course Concepts

The main challenge was **low-texture regions**: without CLAHE, RANSAC failed entirely on data2. Ratio test tuning was also critical — below 0.65 too many true matches were rejected; above 0.80 false matches destabilised the homography.

Course concepts applied: *scale-space / DoG* (SIFT), *gradient orientation histograms* (descriptor design), *homogeneous coordinates & DLT* (homography fitting), *RANSAC* (robust estimation), *projective geometry* (warping), and *Laplacian pyramids* (multi-resolution blending).